MongoDB Data Integrity & Concurrency

Few constraints

PK uniqueness

- Guaranteed by construction
- Single field

Uniqueness – allowed

FK / referential integrity – no

Triggers – no

Transactions

Single document

Isolation levels

Isolation level	Dirty reads	Unrepeatable reads	Phantom reads	Default in
Read uncommitted	Allowed	Allowed	Allowed	
Read committed	Not allowed	Allowed	Allowed	Most, including PostgreSQL
Repeatable read	Not allowed	Not allowed	Allowed	MySQL
Serializable	Not allowed	Not allowed	Not allowed	ANSI standard

ACID vs BASE, CAP?

BASE with an emphasis on CP



Causal consistency

1. Monotonic read
   - Once you see a result, you never see an earlier value
2. Monotonic write
   - Updates are done in the order executed
3. Read-your-writes
   - If you write a record, you always see those updates.
4. Writes-follow-reads
   - Any reads before a write are done before the write



Why use MongoDB?

Data model & querying similar to relational; more flexible!

Weaker consistency increases performance and scalability.